

Cloud Security Fundamentals

THE ARZENS (TAS) — Week 08-09 Combined Assignment — Task 1

What is AWS?

Amazon Web Services (AWS) is a cloud computing platform that rents out computing resources — servers, storage, databases, networking, and dozens of other services — over the internet, billed on a pay-as-you-go basis. Instead of a company buying and maintaining its own physical servers in a data center, it can request virtual resources from AWS in minutes, use them for as long as needed, and stop paying the moment they are no longer needed. AWS operates data centers all over the world, organized into "regions" and "availability zones," which lets customers run their applications physically close to their users and duplicate their systems across locations for resilience.

Why Cloud for Security?

Security teams increasingly rely on cloud platforms like AWS for three main reasons: scalability, cost, and flexibility. **Scalability** means a security tool — for example, a log analysis pipeline — can automatically grow to handle a sudden spike in traffic or shrink back down afterward, without anyone manually installing new hardware. **Cost** savings come from only paying for what is actually used, rather than buying expensive servers that sit mostly idle outside of peak periods. **Flexibility** means new security tools and environments (a test sandbox, a temporary forensics workspace, a new monitoring dashboard) can be spun up and torn down in minutes, which is difficult to do with physical infrastructure. Together, these properties let a small security team operate at a scale that would otherwise require a much larger budget and staff.

Three Core AWS Services

S3 (Simple Storage Service) is object storage: a place to keep files such as log archives, backups, and reports, organized into containers called "buckets." It is durable, cheap, and accessible from anywhere with the right permissions. **EC2 (Elastic Compute Cloud)** provides virtual servers, called "instances," that can run any operating system or application, including security scanners, honeypots, or monitoring agents. Instances can be started and stopped on demand. **IAM (Identity and Access Management)** is the service that controls who — and what — is allowed to do what inside an AWS account. IAM defines users, roles, and permission policies, and it is the backbone of AWS security: almost every security incident in the cloud traces back to a permission that was too broad.

Security Principle: Least Privilege

The principle of **least privilege** states that a user, application, or service should be granted only the exact permissions it needs to do its job — nothing more. For example, a script that only needs to read files from one S3 bucket should never be given permission to delete buckets or manage EC2 instances. Following least privilege limits the damage that can be done if credentials are stolen or a script has a bug, and it is one of the most important habits a security engineer can build when working in AWS.

Simple Flow Diagram



Figure 1. A laptop connects to AWS (authenticated and permission-checked by IAM), which stores the uploaded file in an S3 bucket.